Question

Passing an excess of chlorine gas into a hot concentrated potassium iodide solution, cooling it, and precipitating golden needle-shaped crystals $\mathbf{A}$ (the reaction forming $\mathbf{A}$ is a combination reaction).

Heating $\mathbf{A}$ to constant weight yields a white water-soluble substance $\mathbf{B}$, with a weight loss of 75.8%. Collecting all volatile substances from the decomposition products in a cold trap yields orange-yellow crystals $\mathbf{C}$, but collecting only the less volatile substances at a relatively high temperature yields a dark red liquid, and further cooling yields dark ruby-colored needle-shaped crystals $\mathbf{D}$.

Hint 1: $\mathbf{A}$ is decomposed by water to give a brown solution, while if $\mathbf{A}$ is directly decomposed with an excess of base, a colorless solution is obtained. Write the equations for the reactions under both conditions, respectively.

Hint 2: The actual form in which $\mathbf{C}$ crystals exist is its dimer.

Select the correct option(s) below.

A. The anion of $\mathbf{A}$ and the hybridization of the central atom of molecule $\mathbf{C}$ are different.

B. The anion of $\mathbf{A}$ is a tetrahedral structure.

C. The sum of the reactant coefficients in the ionic equation for the decomposition of $\mathbf{A}$ in water to obtain a brown solution is 14.

D. The sum of the reactant coefficients in the ionic equation for the hydrolysis of $\mathbf{A}$ to obtain a brown solution is 11.

E. The sum of the reactant coefficients in the ionic equation for the decomposition of $\mathbf{A}$ by excess alkali to obtain a colorless solution is 18.

F. The sum of the product coefficients in the ionic equation for the decomposition of $\mathbf{A}$ directly with excess base to obtain a colorless solution is 27.

G. In the ionic equation for the decomposition of $\mathbf{A}$ directly with excess base to obtain a colorless solution, the sum of the reactant coefficients is 21.

H. All of the above statements are incorrect.

Answer

Correct Answer: C

Detailed Explanation

Since KI combines with $\mathrm{Cl_2}$ to form $\mathbf{A}$, the chemical formula of $\mathbf{A}$ can be assumed to be KICl_x

$\mathbf{B}$ is most likely KI or KCl

If $\mathbf{B}$ is KI, then Cl_x is lost,

$35.45x/(39.1 + 35.45x + 126.9) = 75.8\%$, solving for $x = 14.7$, which is discarded

If $\mathbf{B}$ is KCl, then ICl_{x-1} is lost,

$[126.9 + 35.45(x - 1)]/(39.1 + 35.45x + 126.9) = 75.8\%$

Solving for $x = 4.0$, which meets the requirements

Therefore, $\mathbf{A}$ is $\mathrm{KICl_4}$, $\mathbf{B}$ is KCl

CHECKPOINT 2 PTS

$\mathbf{A}$ is $\mathrm{KICl_4}$, $\mathbf{B}$ is KCl

Wherein, in $\mathbf{A}$, I is $\mathrm{sp^3d^2}$ hybridized, and the ion configuration is planar.

CHECKPOINT 0.5 PTS

In $\mathbf{A}$, I is $\mathrm{sp^3d^2}$ hybridized, and the ion configuration is planar

The weight loss is $\mathrm{ICl_3}$, so $\mathbf{C}$ is $\mathrm{ICl_3}$ or written as $\mathrm{I_2Cl_6}$. According to the question, $\mathrm{ICl_3}$ will further decompose when the temperature rises, so $\mathbf{D}$ is ICl.

CHECKPOINT 2 PTS

C is $\mathrm{ICl_3}$ or written as $\mathrm{I_2Cl_6}$, $\mathbf{D}$ is ICl

Wherein, in $\mathrm{I_2Cl_6}$, I is $\mathrm{sp^3d^2}$ hybridized, and two Cl act as bridging ligands, and the molecular configuration is planar.

CHECKPOINT 0.5 PTS

In $\mathrm{I_2Cl_6}$, I is $\mathrm{sp^3d^2}$ hybridized, and two Cl act as bridging ligands, and the molecular configuration is planar.

In prompt 1, $\mathbf{A}$ is decomposed by water to obtain a brown solution. According to the conditions of the question, it is determined that the cause of the brown color is the disproportionation of $\mathbf{A}$ to generate $\mathrm{I_2}$. The ion equation of the reaction is as follows:

$$5\mathrm{ICl_4^-} + 9\mathrm{H_2O} \rightarrow \mathrm{I_2} + 3\mathrm{IO_3^-} + 20\mathrm{Cl^-} + 18\mathrm{H^+}$$

Or

$$5\mathrm{ICl_4^-} + 9\mathrm{H_2O} \rightarrow \mathrm{I_2Cl^-} + 3\mathrm{IO_3^-} + 19\mathrm{Cl^-} + 18\mathrm{H^+}$$

CHECKPOINT 1 PTS

$5ICl_4^- + 9H_2O \to I_2 + 3IO_3^- + 20Cl^- + 18H^+$ Or
$5ICl_4^- + 9H_2O \to I_2Cl^- + 3IO_3^- + 19Cl^- + 18H^+$

If excess alkali is directly used to decompose $\mathbf{A}$ to obtain a colorless solution, it directly disproportionates to generate I^- and IO_3^-. The ion equation of the reaction is as follows:

$$3ICl_4^- + 12OH^- \to I^- + 2IO_3^- + 12Cl^- + 6H_2O$$

CHECKPOINT 1 PTS

$3ICl_4^- + 12OH^- \to I^- + 2IO_3^- + 12Cl^- + 6H_2O$

In summary, option C is correct.